PKM fall-degree study, stage A relaunch (pfd-a-20261108): results

Job pfd-a-20261108 (pod 2317938f-ba8b-4d59-a081-1e9ece2ae42f), 2026-11-08 17:14-20:47Z, 12 vCPU, RAYON_NUM_THREADS 12, r6i-class node (request pins r6i.4xlarge; host.txt: Xeon Platinum 8375C, 129.8 GB). Image digest sha256:0d6b2ecd..., source f13a690df0. Protocol: research/prime_fall_degree_20261027/PREREGISTRATION.md sha256 7bfc1dec... (Amendments 1-9). Every number is [pcc:unverified]; values are measured (read from the job's rows) or derived (by the scratchpad scorer).

Verdict

Stage A reproduces: the decision-1 median-level repeat check agrees at every t <= 10, so Amendment 2 does not stop the study. C-step (Amendment 6): holds at t = 11 (not a blind test, Amendment 7 E); UNDECIDED at the three blind cells t = 12, 13, 14 (median d_F >= 6 everywhere; the bound includes the predicted 6, 6, 7). Neither confirmed nor refuted. Kummer m = 2 at P1: H-LFD 'd_F grows', H-FFD 'FFD assumption fails', H-gap 0 of 22 draws with d_F < D. H-null: UNRESOLVED by the rule text (two largest common N with rows, t = 13, 14, are censored on both arms; on completed N, t = 9, 10, it reads 'shape-driven').

Gates

- G1: 6 cases pass (hky_fp_tower_check; the registered 'cargo test --lib hky_fp_tower' runs 0 tests, RESTART.md).
- G2 (d_F <= D): 72 draws checked (52 in the registered Kummer/isogeny cells), 0 violations.
- G3: 144 capped runs, 0 disagreements; 72 systems x caps d-1 and d, N <= 20, 0 seed mismatches, 0 timeouts. The tool tests 2 caps per system, not every cap from 2 to d_F (252 runs) as section 4 reads.
- G4: not part of this job (it is in pfd-c); Amendment 3 records its first-run pass (records lost).

Repeat check (owner decision 1, median level)

Run medians t = 1..10: 3, 4, 4, 4, 4, 5, 5, 5, 5, 6; reference 3,4,4,4,4,5,5,5,5,6; AGREE at every t (both draws equal the median at every t).

C-step at t >= 11 (Amendment 7 D.5)

- t = 11: c* = 6, median d_F 6, holds at t (not blind); source: =D (F4); ffd = 4.
- t = 12: c* = 6, median d_F >=6, undecided at t (bound includes c*); source: bracket incomplete: cap 5 not refuting, cap 6 not run -> d_F >= 6; ffd = 4.
- t = 13: c* = 6, median d_F >=6, undecided at t (bound includes c*); source: closure one-sided; ffd = 4.
- t = 14: c* = 7, median d_F >=6, undecided at t (bound includes c*); source: closure one-sided; ffd = 4.

Missing for a decision: the Amendment 7 B cap-6 runs at t = 12 and 13 and the cap-6/cap-7 runs at t = 14 (pfd-x-* jobs, not part of pfd-a). Uncapped F4 at t = 12 stopped at its 1800 s budget (censored); cap-A ran only cap 5 (D - 1 from the timed-out D).

Controls

Null (random coefficients on the S3 support): identical d_F, ffd and D to the Kummer arm on every draw at t <= 10, and identical D at t = 11. Naive: d_F 4, 6 at t = 1, 2 and > 7 at t = 3..6 (closure to cmax 7 without 1), consistent with the proved bound d_F >= 2^t (DERIVED D5). Decomposability (Amendment 7 C): grid_zeros = 0 on all 44 Kummer tower draws (pfd_grid_zeros; cross-checked by ns-degree to t = 12 and a numpy recount to t = 14), so nothing is excluded.

Scorer dependence

The Amendment 7 D scorer patch is not done; a scratchpad scorer written from the rule text was used. Only the t = 11 value depends on it (=D (F4): 6; the committed scorer prints the closure's one-sided >= 6). No reading changes: H-LFD and H-FFD read the same on t <= 10 alone, and t = 11 is not a blind C-step cell. The committed scorer (dev build of the same source) reproduces every other median and both ladder readings.

Caveats

- 2 draws per cell; equal draws at every t. No timing claim; wall seconds are host-dependent. F4 muladds are compared only within this 12-thread job.
- The instance type is inferred (request pin, CPU model, memory); host.txt does not name it.
- t = 12 CAP rows ran under run.sh cap-A limits (--budget 1800, --max-nnz), not Amendment 7 A's 86,400 s/--max-dense; they finished uncensored in 17 s.

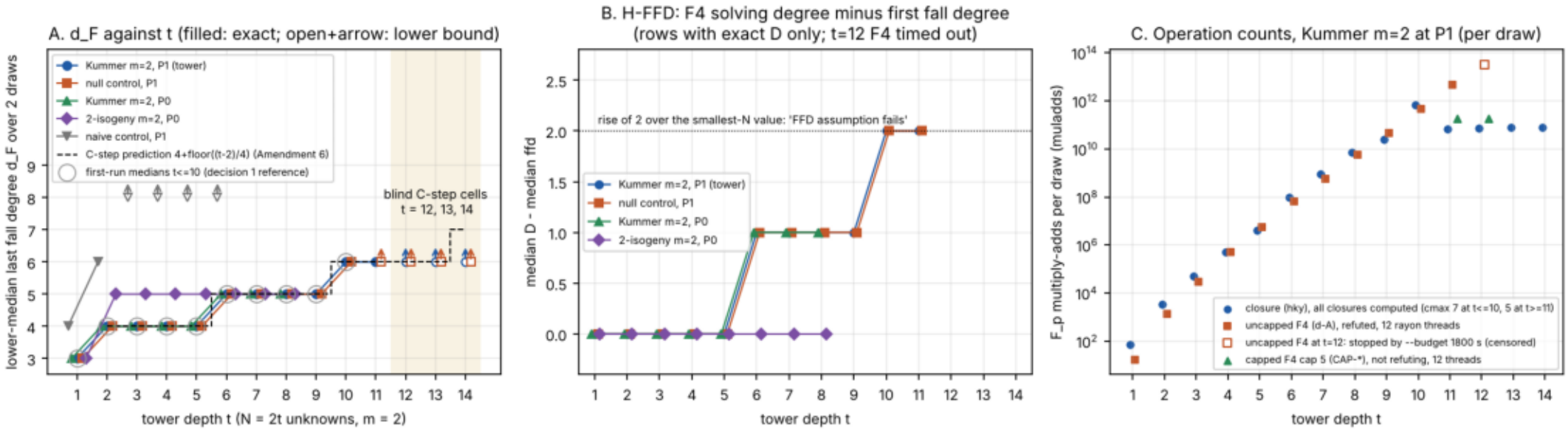


Figure 1. A: lower-median d_F against t per arm, the C-step prediction and the decision-1 reference. B: H-FFD series. C: F_p multiply-adds per draw (closure and F4 at 12 threads). Source rows: results/R9/2317938f-.../runs/. [pcc:unverified]